TidyTuesday Week 42

Pranay Gundam

Wednesday 14th October, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly pilgrimage through the economic junk drawer, where we grab two time series that have never met, force them into a regression, and act shocked when they have opinions about each other. This week delivered a genuinely rare treat, a magnificent fraud, and one of the most beautifully plausible-looking lies the project has ever produced. Strap in.

Monday kicked us off with the Balance on Services (a discontinued U.S. trade statistic, because of course it is) versus the exchange rate of the Belize dollar per U.S. dollar. The raw regression squeaked out significance with a p-value of 0.045, which is the statistical equivalent of a result clearing the bar by tripping over it. But here's the twist nobody ordered: once we detrended both series, the relationship didn't collapse – it actually got stronger, p dropping to 0.018 and even flipping sign. So after stripping out the shared march of time, U.S. services trade and the Belizean exchange rate still want to hold hands. Is there a deep economic story here? Almost certainly not. But this is the rarer, more interesting case the project pretends to care about, so we will dramatically raise one eyebrow and move on.

Tuesday was the showstopper. We paired average gasoline prices in the South Census Region, Size Class B (a level of specificity I find genuinely moving) against the 30-Year FHA Mortgage Rate. The raw regression was strong, R-squared around 0.17, p-value in the land of absurdly small. And then the detrended version didn't just survive – it absolutely flexed, with an R-squared of 0.63 and a p-value with fifty-one zeros after the decimal point. Fifty-one. That is not a correlation, that is a marriage. Both gas prices and mortgage rates respond to the same underlying inflationary mood of the late 20th century, so there's an actual thread of economic sense here, which is deeply unsettling for a project built on nonsense. I may need to lie down.

Wednesday restored cosmic balance by handing us the textbook trend-mirage. Corporate machinery-sector cash holdings versus the debt liabilities of Government-Sponsored Enterprises posted a raw R-squared of 0.80 – gorgeous, dazzling, the kind of number that gets you invited to parties. Then detrending took it out behind the shed: R-squared cratered to 0.02 and the p-value ballooned to 0.13, officially Not Significant and vaguely embarrassed. Both series just drift upward over the decades because everything denominated in dollars drifts upward over the decades, and once you remove that shared escalator, the "relationship" evaporates like my motivation on a Friday. This is exactly the kind of result this project lives to dunk on.

As always, the mandatory reminder: these pairs are chosen at random, nobody is claiming gasoline causes mortgages or that Belize is secretly steering American trade, and spurious correlation is the house specialty, not the exception. Two of this week's three "findings" either held up

or imploded in exactly the ways statistics textbooks warn you about, and that's the whole dumb point. See you next week for more.

2 Date: 2026-10-12

Series ID: BOPBSVN
This series is titled Balance on Services (DISCONTINUED) and has a frequency of Quarterly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1960-01-01 and the observation end date is 2014-01-01.The popularity of this series is 1.

Series ID: XRNCUSBZA618NRUG
This series is titled Exchange Rate (market+estimated) for Belize and has a frequency of Annual. The units are National Currency Units per US Dollar and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1970-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG	**R-squared:**	0.090
Model:	OLS	**Adj. R-squared:**	0.069
Method:	Least Squares	**F-statistic:**	4.246
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0454
Time:	20:28:58	**Log-Likelihood:**	34.026
No. Observations:	45	**AIC:**	-64.05
Df Residuals:	43	**BIC:**	-60.44
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9274	0.024	79.937	0.000	1.879	1.976
value_fred_BOPBSVN	0.0023	0.001	2.060	0.045	4.83e-05	0.004

Omnibus:	17.591	**Durbin-Watson:**	0.425
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	23.370
Skew:	-1.275	**Prob(JB):**	8.42e-06
Kurtosis:	5.441	**Cond. No.**	30.5

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

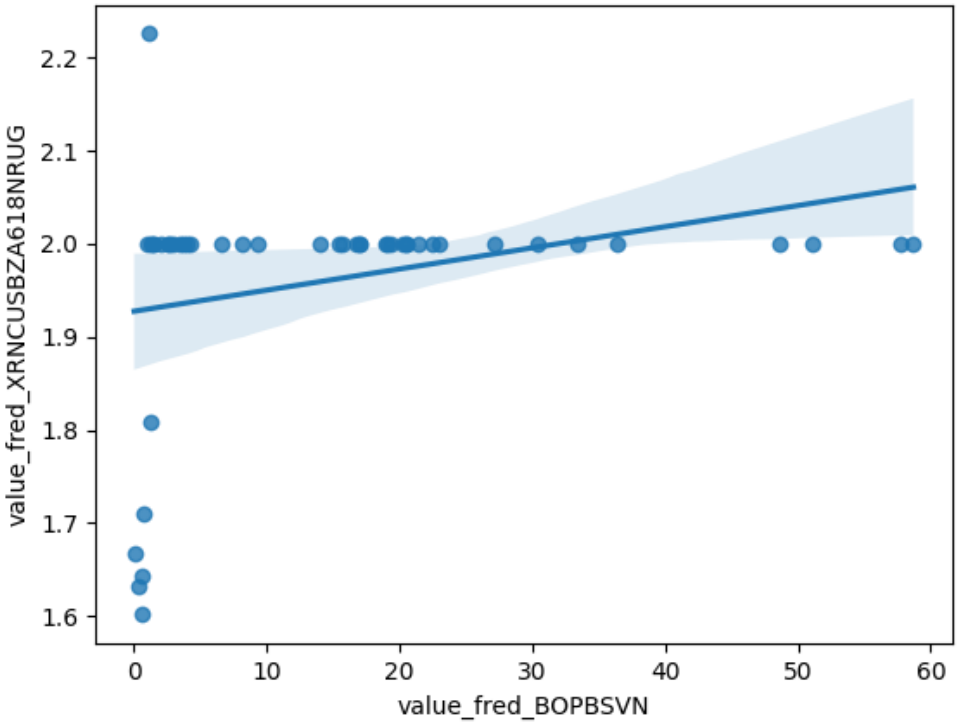


Figure 1: Raw Regression Plot for 2026-10-12

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG_detrended	**R-squared:**	0.123
Model:	OLS	**Adj. R-squared:**	0.103
Method:	Least Squares	**F-statistic:**	6.041
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0181
Time:	20:28:58	**Log-Likelihood:**	41.263
No. Observations:	45	**AIC:**	-78.53
Df Residuals:	43	**BIC:**	-74.91
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.671e-15	0.015	3.84e-13	1.000	-0.030	0.030
value_fred_BOPBSVN_detrended	-0.0050	0.002	-2.458	0.018	-0.009	-0.001

Omnibus:	8.915	**Durbin-Watson:**	0.588
Prob(Omnibus):	0.012	**Jarque-Bera (JB):**	16.952
Skew:	0.296	**Prob(JB):**	0.000208
Kurtosis:	5.948	**Cond. No.**	7.28

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

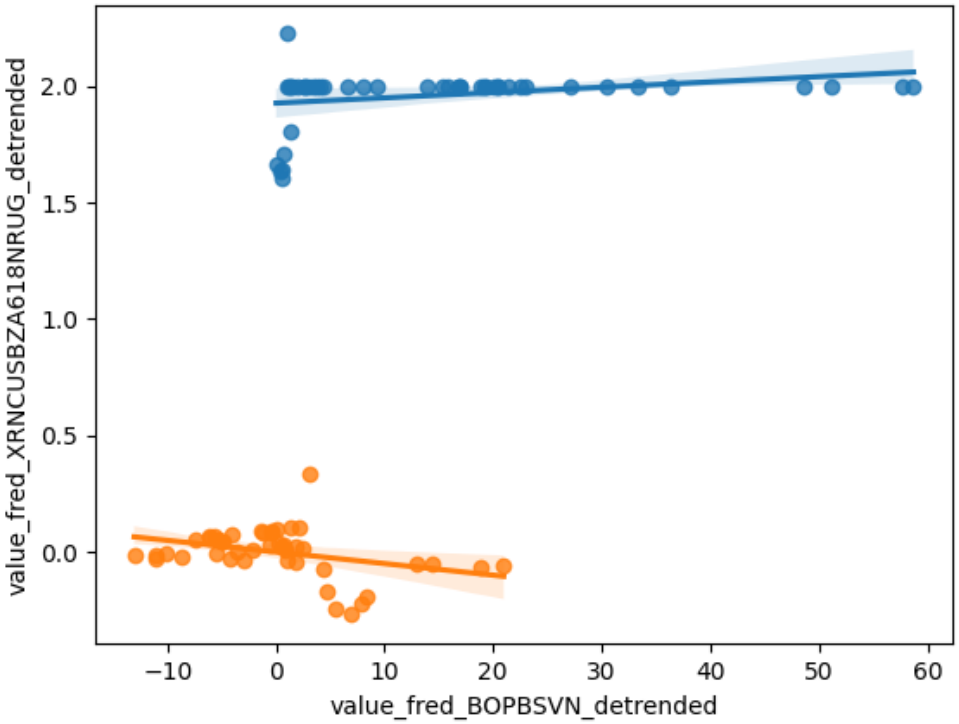


Figure 2: Detrended Regression Plot for 2026-10-12

3 Date: 2026-10-13

Series ID: APUB30074714
This series is titled Average Price: Gasoline, Unleaded Regular (Cost per Gallon/3.785 Liters) in the South Census Region - Size Class B and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1978-01-01 and the observation end date is 1997-12-01.The popularity of this series is 1.

Series ID: FHA30
This series is titled 30-Year FHA Mortgage Rate: Secondary Market (DISCONTINUED) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1964-01-01 and the observation end date is 2000-06-01.The popularity of this series is 9.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FHA30	**R-squared:**	0.169
Model:	OLS	**Adj. R-squared:**	0.165
Method:	Least Squares	**F-statistic:**	47.29
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	5.59e-11
Time:	18:35:32	**Log-Likelihood:**	-534.79
No. Observations:	235	**AIC:**	1074.
Df Residuals:	233	**BIC:**	1081.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	3.5713	1.048	3.408	0.001	1.507	5.636
value_fred_APUB30074714	6.5135	0.947	6.876	0.000	4.647	8.380

Omnibus:	11.293	**Durbin-Watson:**	0.039
Prob(Omnibus):	0.004	**Jarque-Bera (JB):**	6.092
Skew:	0.195	**Prob(JB):**	0.0476
Kurtosis:	2.315	**Cond. No.**	13.6

Notes:
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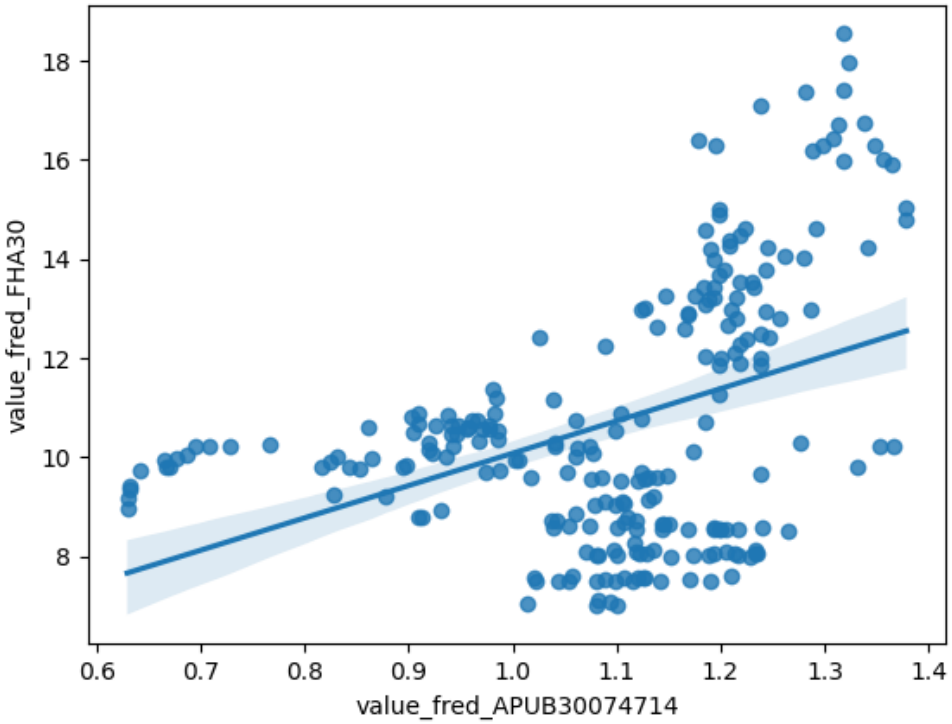


Figure 3: Raw Regression Plot for 2026-10-13

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FHA30_detrended	**R-squared:**	0.626
Model:	OLS	**Adj. R-squared:**	0.624
Method:	Least Squares	**F-statistic:**	389.6
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	1.23e-51
Time:	18:35:32	**Log-Likelihood:**	-352.60
No. Observations:	235	**AIC:**	709.2
Df Residuals:	233	**BIC:**	716.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.561e-13	0.071	7.82e-12	1.000	-0.140	0.140
value_fred_APUB30074714_detrended	8.7393	0.443	19.738	0.000	7.867	9.612

Omnibus:	5.577	**Durbin-Watson:**	0.209
Prob(Omnibus):	0.062	**Jarque-Bera (JB):**	5.236
Skew:	0.334	**Prob(JB):**	0.0729
Kurtosis:	3.299	**Cond. No.**	6.23

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

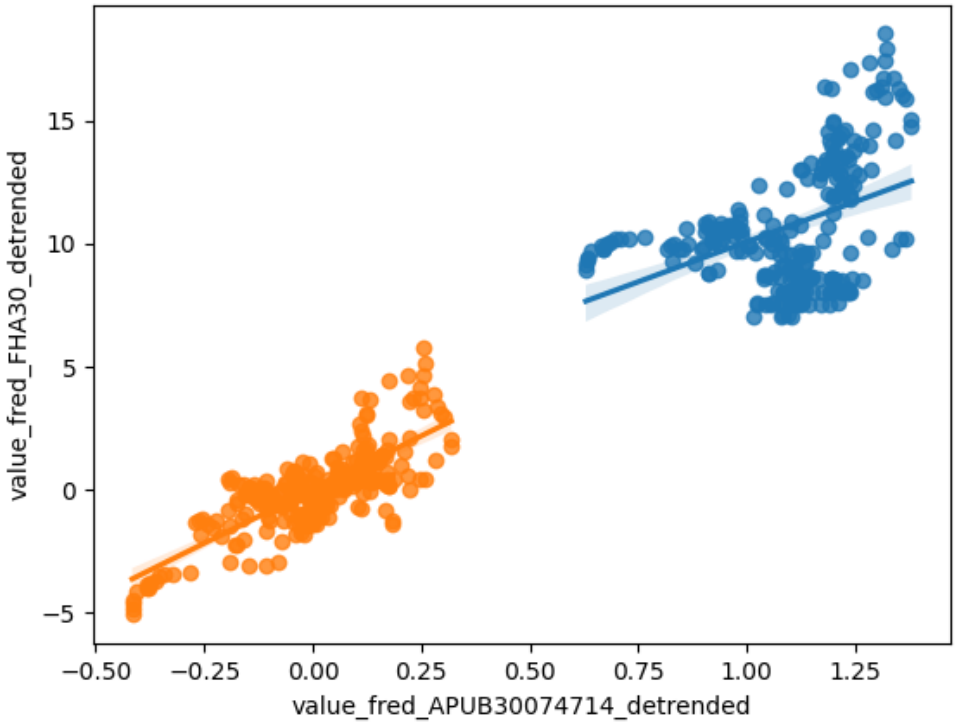


Figure 4: Detrended Regression Plot for 2026-10-13

4 Date: 2026-10-14

Series ID: QFR201333USNO
This series is titled Quarterly Financial Report: U.S. Corporations: Machinery: Cash and Demand Deposits in the U.S. and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-10-01 and the observation end date is 2026-04-01.The popularity of this series is 1.

Series ID: GSETCMDODFS
This series is titled Government-Sponsored Enterprises; Debt Securities and Loans; Liability, Level and has a frequency of Quarterly, End of Period. The units are Millions of U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1945-10-01 and the observation end date is 2026-04-01.The popularity of this series is 2.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_GSETCMDODFS	**R-squared:**	0.805
Model:	OLS	**Adj. R-squared:**	0.803
Method:	Least Squares	**F-statistic:**	416.8
Date:	Wed, 14 Oct 2026	**Prob (F-statistic):**	1.26e-37
Time:	16:02:51	**Log-Likelihood:**	-1578.2
No. Observations:	103	**AIC:**	3160.
Df Residuals:	101	**BIC:**	3166.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.816e+05	2.74e+05	1.758	0.082	-6.18e+04	1.02e+06
value_fred_QFR201333USNO	237.3381	11.626	20.415	0.000	214.276	260.400

Omnibus:	4.636	**Durbin-Watson:**	0.398
Prob(Omnibus):	0.098	**Jarque-Bera (JB):**	4.489
Skew:	-0.511	**Prob(JB):**	0.106
Kurtosis:	2.949	**Cond. No.**	5.94e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.94e+04. This might indicate that there are strong multicollinearity or other numerical problems.

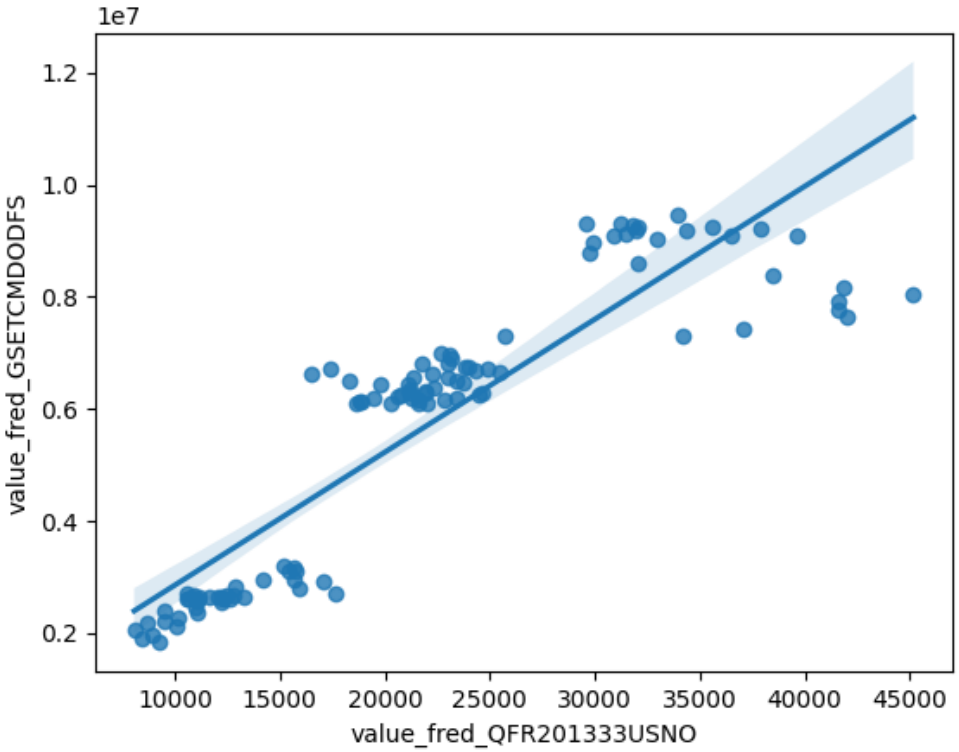


Figure 5: Raw Regression Plot for 2026-10-14

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_GSETCMDODFS_detrended	**R-squared:**	0.022
Model:	OLS	**Adj. R-squared:**	0.013
Method:	Least Squares	**F-statistic:**	2.317
Date:	Wed, 14 Oct 2026	**Prob (F-statistic):**	0.131
Time:	16:02:51	**Log-Likelihood:**	-1530.9
No. Observations:	103	**AIC:**	3066.
Df Residuals:	101	**BIC:**	3071.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-4.732e-09	6.86e+04	-6.9e-14	1.000	-1.36e+05	1.36e+05
value_fred_QFR201333USNO_detrended	28.1415	18.489	1.522	0.131	-8.536	64.819

Omnibus:	13.571	**Durbin-Watson:**	0.344
Prob(Omnibus):	0.001	**Jarque-Bera (JB):**	16.691
Skew:	0.705	**Prob(JB):**	0.000237
Kurtosis:	4.378	**Cond. No.**	3.71e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.71e+03. This might indicate that there are strong multicollinearity or other numerical problems.

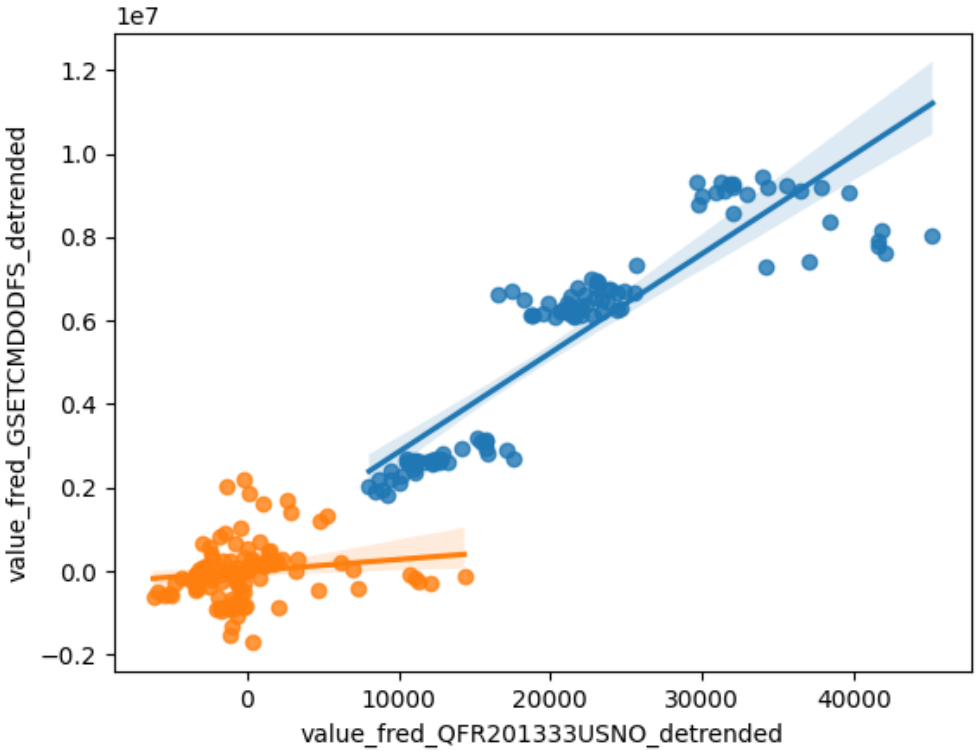


Figure 6: Detrended Regression Plot for 2026-10-14